

Mathematical and Computational Methods

Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

Tutorial route (►, in this order, about 80 minutes): 1.3, 2.4, 2.5(a,b), 2.6, 2.8, 2.10(a,b), 2.3. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 (★ drill — exponential form). Write each number in exponential form $r e^{i\theta}$ with $r > 0$ and $-\pi < \theta \leq \pi$:

(a) $1 + i$, (b) -3 , (c) $2i$, (d) $-1 - i$, (e) $\sqrt{3} - i$, (f) $-2 + 2\sqrt{3}i$.

Exercise 1.2 (★ drill — back to Cartesian; Euler's identity). Write each in Cartesian form $a + ib$:

(a) $2 e^{i\pi/3}$, (b) $3 e^{i\pi}$, (c) $\sqrt{2} e^{-i\pi/4}$, (d) $4 e^{i3\pi/2}$, (e) $e^{i5\pi/6}$, (f) $e^{i2\pi}$.

(g) Use (b) to state Euler's identity.

Exercise 1.3 (► ★ drill — rotation). Let $z = 2 e^{i\pi/6}$. Write down $e^{i\pi/2} z$ in exponential form, and describe geometrically the effect of multiplying a complex number by (i) i , (ii) $e^{i\pi}$, (iii) $e^{-i\pi/2}$.

Exercise 1.4 (★ drill — integer powers). Compute, using the exponential form:

(a) $(1 + i)^4$, (b) $(\sqrt{3} + i)^3$, (c) $(1 - i)^6$, (d) $(2 e^{i\pi/5})^{10}$.

(e) Use De Moivre's theorem to simplify $(\cos \theta + i \sin \theta)^{-2}$.

Exercise 1.5 (★ drill — a few roots). Find all the (a) square roots of i and (b) fourth roots of 1 . Plot the roots in (b) on an Argand diagram, mark the unit circle, and name the polygon they form.

Exercise 1.6 (★ drill — the fundamental theorem of algebra). (a) State the fundamental theorem of algebra, and say what “counted with multiplicity” means. (b) How many roots in $\mathbb{C}$, counted with multiplicity, has (i) a degree-5 polynomial; (ii) $z^7 - 3 = 0$; (iii) $(z - 1)^2(z^2 + 1) = 0$? (c) True or false: every real cubic has at least one real root?

Exercise 1.7 (★ drill — hyperbolic link). Using $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, evaluate $\cos(i)$ in terms of e . (The general identity $\cos(ix) = \cosh x$ is built up in Exercise 2.9.)

Exercise 1.8 (★ drill — phasors). (a) Write $5 \cos(\omega t + \frac{\pi}{3})$ and $2 \sin \omega t$ as phasors $P = A e^{i\varphi}$. (b) The phasor $3 e^{-i\pi/2}$ represents which real signal $A \cos(\omega t + \varphi)$?

2 Tutorial problems

Exercise 2.1 (★★core — arithmetic in exponential form). Let $z = \sqrt{3} + i$ and $w = 1 + i$. Write z and w in exponential form, then find zw and z/w in both exponential and Cartesian form.

Exercise 2.2 (★★core — rotation in action). (a) Rotate $z = 3 + 4i$ anticlockwise by $\frac{\pi}{2}$ about the origin, and then (separately) clockwise by $\frac{\pi}{2}$. (b) Rotate $2 + 2i$ anticlockwise by $\frac{\pi}{4}$. Give exact Cartesian answers.

Exercise 2.3 (► ★★core — prove De Moivre (stretch)). Prove by induction that $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for all integers $n \geq 1$. (Pure-maths.)

Exercise 2.4 (► ★★core — powers by De Moivre (anchor)). Compute (a) $(1 + i)^8$ and (b) $(\sqrt{3} + i)^6$ using De Moivre's theorem.

Exercise 2.5 (► ★★core — roots, two shapes (anchor)). Find (a) the cube roots of 8 and (b) the fourth roots of -16 . In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

Exercise 2.6 (► ★★core — sixth roots of unity (anchor)). Find all sixth roots of unity in Cartesian form, and verify that they sum to zero. Why does this generalise to all $n \geq 2$?

Exercise 2.7 (★★core — data-science flavour: twiddle factors). The fast Fourier transform is built from the “twiddle factors” $W_N^k = e^{-i2\pi k/N}$. Take $N = 8$. (a) Write W_8^1 in Cartesian form. (b) Show that $W_8^{k+4} = -W_8^k$ for every integer k (the symmetry that halves the FFT's work). (c) Give the modulus and argument of W_8^3 .

Exercise 2.8 (► ★★core — $\cos 3\theta$ and $\sin 3\theta$ (anchor)). Using De Moivre and the binomial theorem, show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.

Exercise 2.9 (★★core — trig-hyperbolic link). (a) Show $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$. (b) Hence evaluate $\cos(i \ln 2)$ exactly.

Exercise 2.10 (► ★★core — combining oscillations (anchor; physics)). (a) Combine $3 \cos \omega t + 4 \sin \omega t$ into a single sinusoid $R \cos(\omega t - \alpha)$. (b) Combine $2 \cos \omega t + 2 \cos(\omega t + \frac{\pi}{2})$ likewise.

3 Further practice (home)

Roots and the fundamental theorem (Skills 5–6)

Exercise 3.1 (★★core — solving $w^n = z$ with a non-real right-hand side). (a) Solve $w^3 = -27i$, giving the three roots in exponential and Cartesian form. (b) Solve $w^3 = 1 + i$, leaving the roots in exponential form, and check that each root is the previous one rotated through $\frac{2\pi}{3}$ about the origin.

Exercise 3.2 (★★core — square root the algebraic way). The argument of $5 - 12i$ is not a standard angle, so the exponential method is awkward. Instead find both square roots of $5 - 12i$ in Cartesian form by equating real and imaginary parts of $(a + ib)^2$, and verify by squaring.

Exercise 3.3 (★★core — FTA and real factors). (a) Verify that $z = 2i$ is a root of $z^2 + 4$, and write down the other. (b) Factor $z^4 + 4$ into two real quadratics. [Hint: find the four roots first.]

Exercise 3.4 (★★★challenge — a conjugate equation). Find all $z \in \mathbb{C}$ with $z^2 = \bar{z}$.

Trigonometric identities (Skill 7)

Exercise 3.5 (★★core — power reduction). Use the inverse Euler formula to write $\sin^4 \theta$ as a sum of cosines of multiples of θ .

Exercise 3.6 (★★★challenge — a finite cosine sum). Using the geometric series for $\sum_{k=0}^{n-1} e^{ik\theta}$ (with $\theta \neq 0 \bmod 2\pi$), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{(n-1)\theta}{2}\right).$$

Trigonometric–hyperbolic link (Skill 8)

Exercise 3.7 (★★core — the converse pair). Show that $\cosh(ix) = \cos x$ and $\sinh(ix) = i \sin x$, and evaluate $\sin(2i)$ and $\cos(\pi i)$ in terms of e .

Phasors (Skill 9)

Exercise 3.8 (★★core — physics word problem: two AC sources). Two alternating voltages of equal amplitude but different phase, $V_1 = 10 \cos \omega t$ and $V_2 = 10 \cos(\omega t + \frac{2\pi}{3})$ (volts), are added. Find the amplitude and phase of $V_1 + V_2$.